Economic Impact of Video Games: A Big Data Approach Using Global Sales Data

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Context & Objectives

- Video games evolved into a multi-billion dollar global industry
- Traditional analysis struggles with scale, heterogeneity, and temporal depth
- Objective: **design and validate a Big Data pipeline** to analyze long-term industry trends

Literature Review

- Prior studies focus on:
  - Console life cycles
  - Genre dominance
  - Regional markets
- Industry reports (Newzoo, Visual Capitalist) show revenue growth but lack transparent pipelines
- **Gap:** Limited reproducible, scalable analytical pipelines

Dataset Overview

- Source: Video Game Sales (1978–2024)
- Records: ~70k
- Key attributes:
  - Year, Platform, Genre
  - Regional sales (NA, PAL, JP, Other)
  - Global sales
- Physical sales bias
- No mobile-first revenue

Apache Spark and Colab

- Distributed processing
- Scales beyond pandas
- Built-in aggregation & validation

- Reproducible
- No local environment issues
- Adequate for batch analytics

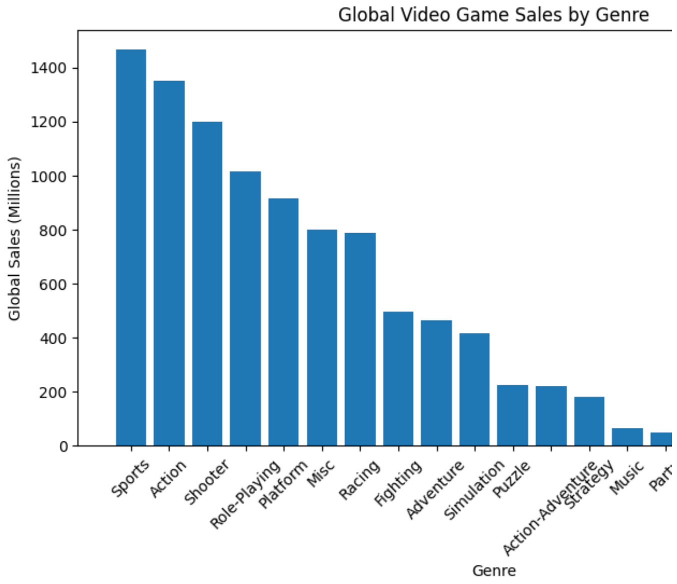
Lakehouse Architecture

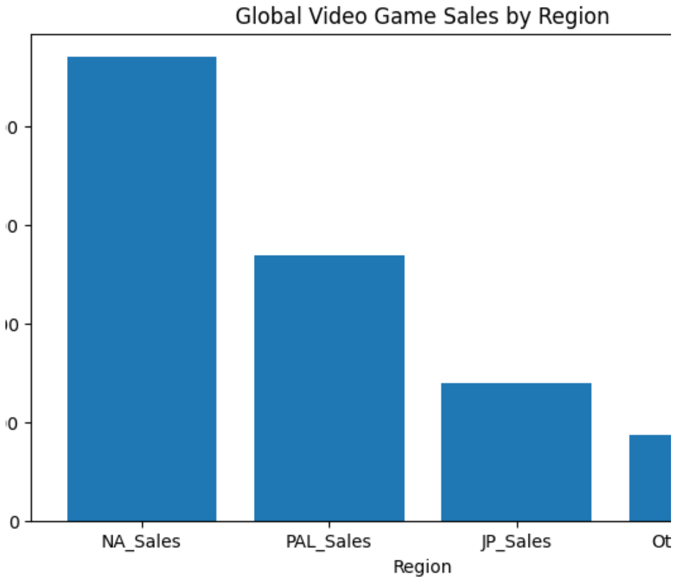
Validation Strategy

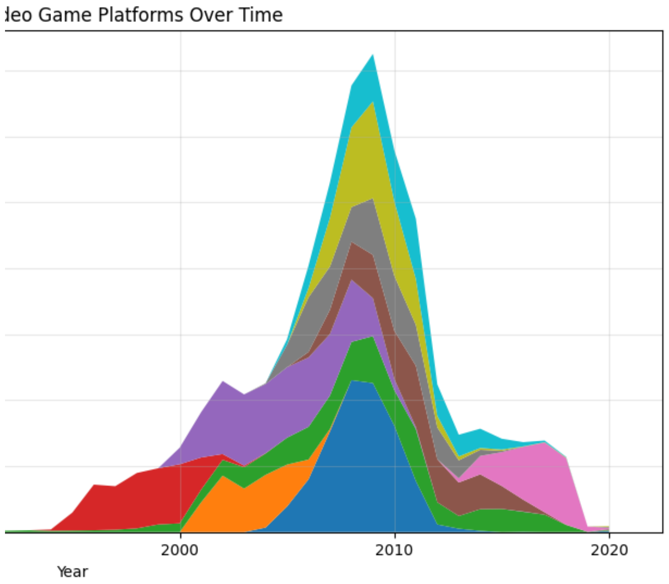
- Incremental validation
- Hard Rules
- Soft Rules
- No over-cleaning
- Regional vs global sales consistency
- Duplicate Removal
- Platform Normalization

Results: Global Trends

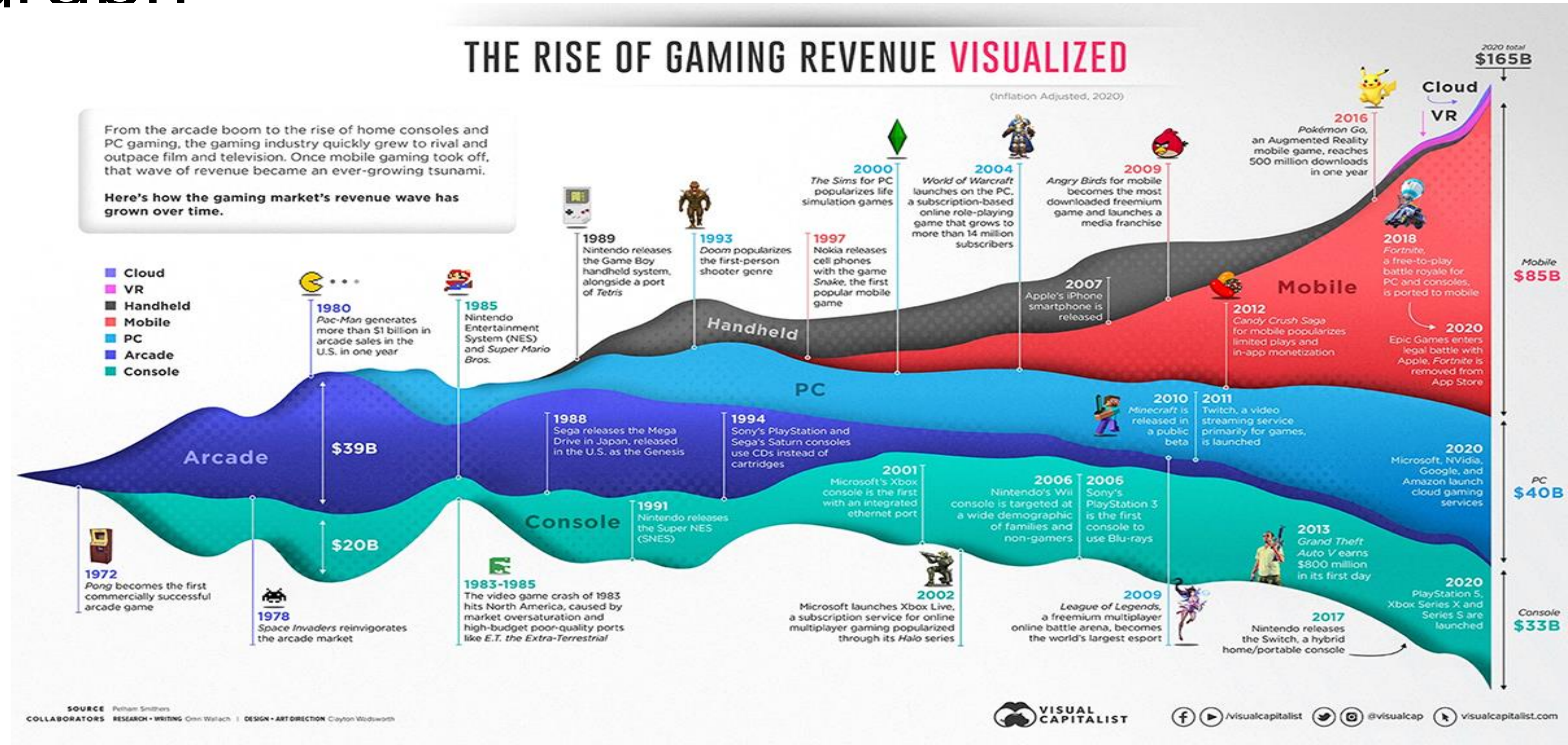
Platform & Genre Dynamics







Comparison with External Industry Graph



Limitations & Trade-offs

- No inflation adjustment
- Physical sales bias
- Platform metadata inconsistency

Thank you